

1. What is JavaScript?

JavaScript is a programming language used to make web pages interactive.

It can:

- Handle user actions
- Update content dynamically
- Validate forms
- Create animations

Runs in browser (client-side)

2. JavaScript Syntax

```
let x = 10;
```

Explanation:

- `let` → keyword to declare variable
- `x` → variable name
- `10` → value

Use:

Store data

3. Variables

```
var a = 5;  
let b = 10;  
const c = 15;
```

Explanation:

- var → old method
- let → modern variable
- const → constant (cannot change)

Use:

Store values

4. Data Types

```
let name = "John"; // String  
let age = 20;      // Number  
let isStudent = true; // Boolean
```

Types:

- String
- Number
- Boolean
- Object
- Undefined
- Null

Use:

Define type of data

5. Operators

```
let sum = 5 + 3;
```

Types:

- Arithmetic → + - * /
- Comparison → == === !=
- Logical → && || !

Use:

Perform operations

6. Functions

```
function greet() {  
  console.log("Hello");  
}
```

Explanation:

- function → defines function
- greet() → function name

Use:

Reusable code

7. Arrow Function

```
const greet = () => {  
  console.log("Hello");  
};
```

Use:

Short syntax

8. Conditional Statements

```
if (age > 18) {  
  console.log("Adult");  
} else {  
  console.log("Child");  
}
```

Use:

Decision making

9. Switch Statement

```
switch(day) {  
  case 1: console.log("Monday"); break;  
}
```

Use:

Multiple conditions

10. Loops

For Loop

```
for(let i=0; i<5; i++) {  
  console.log(i);  
}
```

While Loop

```
while(i < 5) {  
  i++;  
}
```

Use:

Repeat code

11. Arrays

```
let arr = [1,2,3];
```

Methods:

```
arr.push(4);  
arr.pop();
```

Use:

Store multiple values

12. Objects

```
let person = {  
  name: "John",  
  age: 20  
};
```

Use:

Store structured data

13. DOM (Document Object Model)

```
document.getElementById("id").innerHTML = "Hello";
```

Use:

Change HTML content

14. Events

```
button.onclick = function() {  
  alert("Clicked");  
};
```

Use:

Handle user actions

15. Event Listeners

```
button.addEventListener("click", function() {  
  alert("Clicked");  
});
```

Use:

Better event handling

16. Form Validation

```
if(name == "") {  
  alert("Enter name");  
}
```

Use:

Check user input

17. Strings

```
let text = "Hello";  
text.length;  
text.toUpperCase();
```

Use:

Text operations

18. Numbers

```
let num = 10;  
num.toString();
```

Use:

Number operations

19. Math Object

```
Math.random();  
Math.round(4.5);
```

Use:

Calculations

20. Date Object

```
let date = new Date();
```

Use:

Work with dates

21. SetTimeout & setInterval

```
setTimeout(() => {  
  alert("Hello");  
}, 2000);
```

Use:

Delay execution

22. JSON

```
let obj = JSON.parse('{"name":"John"}');
```

Use:

Data exchange

23. Local Storage

```
localStorage.setItem("name", "John");
```

Use:

Store data in browser

24. ES6 Features

```
let x = 10;  
const y = 20;
```

Includes:

- Arrow functions
- Template literals
- Destructuring

Use:

Modern JS

25. Template Literals

```
let name = "John";  
console.log(`Hello ${name}`);
```

Use:

Dynamic strings

26. Destructuring

```
let {name, age} = person;
```

Use:

Extract values

27. Spread Operator

```
let newArr = [...arr];
```

Use:

Copy arrays

28. Promises

```
let promise = new Promise((resolve, reject) => {  
  resolve("Done");  
});
```

Use:

Handle async tasks

29. Async/Await

```
async function test() {  
  let result = await promise;  
}
```

Use:

Simplify async code

30. Fetch API

```
fetch("url")  
.then(res => res.json())  
.then(data => console.log(data));
```

Use:

Get data from server

31. Error Handling

```
try {  
  let x = y;  
} catch(e) {  
  console.log("Error");  
}
```

Use:

Handle errors

32. Classes

```
class Person {  
  constructor(name) {  
    this.name = name;  
  }  
}
```

Use:

OOP

33. Modules

```
export default function test() {}
```

Use:

Code organization